

A High-School Friendly Tutorial On the Cell2location Bayesian Model

Explaining the model shown on page 12 of
Cell2location maps fine-grained cell types in spatial transcriptomics

This tutorial explains the page-12 Bayesian model step by step: what is observed, what is unknown, what is being modeled, what the symbols mean, why priors are used, which prior distributions are used for the main parameters, and why the user-adjusted hyperparameters matter.

Prepared as a learning guide for a high school student
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1 What problem is cell2location trying to solve?

Imagine a thin slice of tissue, such as a slice of mouse brain or a human lymph node. A spatial transcriptomics machine measures many small locations on the tissue. In a common technology such as 10x Visium, each location, often called a **spot**, can contain RNA from more than one cell.

For each spot, the machine gives a table of counts:

Spot	Gene 1	Gene 2	Gene 3	Gene 4
Spot A	18	0	43	6
Spot B	2	31	7	20
Spot C	15	12	10	4

These counts are not direct labels like “this spot has 3 T cells and 1 B cell.” They are just gene-expression counts. The job of cell2location is to use those gene counts to estimate:

How much of each cell type is present at each spot?

Big idea

Cell2location is a Bayesian model for turning gene counts at tissue locations into estimated cell-type abundances. It uses reference cell-type signatures from single-cell or single-nucleus RNA-seq, then asks: “Which mixture of reference cell types most likely produced the observed gene counts at this spot?”

1.1 Why do we need a model at all?

If every spot contained exactly one cell, and every cell type had one perfect marker gene, the problem would be easy. But real data are messy:

- A spot may contain several cells.
- Different cell types can express many of the same genes.
- Some cell types are very similar to each other.
- The machine may capture more RNA in some spots than in others.
- Some RNA floating around in the experiment may contaminate a spot.
- Counts are noisy. Repeating the same experiment would not give exactly the same numbers.

A model is a careful mathematical story for how the observed counts could have been generated. Bayesian inference then runs this story backward: from observed counts to likely hidden cell-type abundances.

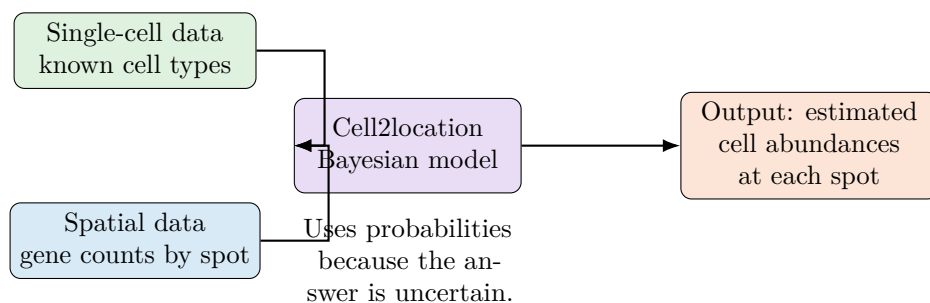


Figure 1: A simplified workflow. Single-cell data tell us what each cell type usually expresses. Spatial data tell us what was observed at each spot. Cell2location combines them.

2 A tiny biology and data dictionary

You do not need to be a biologist to understand the model, but these words help.

Term	Plain-English meaning
Gene	A DNA instruction. In this model, we only care about how much RNA from a gene is detected.
mRNA	Messenger RNA. More mRNA from a gene usually means that gene is more active.
RNA count	The number reported by the sequencing technology for a gene in a spot. Counts are nonnegative whole numbers: 0, 1, 2, and so on.
Cell type	A category of cell, such as T cell, B cell, astrocyte, or neuron subtype.
Reference signature	A typical gene-expression pattern for one cell type. It says things like: “T cells usually have high expression of gene A and low expression of gene B.”
Spatial spot/location	A small location on the tissue slide. It may contain several cells or parts of cells.
Batch/experiment	A slide, sample, or technology run. Two batches can have different technical behavior.

3 Bayesian inference in one page

The whole Bayesian idea can be written as:

Model piece

$$\underbrace{p(\text{unknowns} \mid \text{data})}_{\text{posterior}} \propto \underbrace{p(\text{data} \mid \text{unknowns})}_{\text{likelihood}} \times \underbrace{p(\text{unknowns})}_{\text{prior}}.$$

Read it like this:

- **Unknowns:** things we want to learn, such as how many cells of each type are in each spot.
- **Data:** the gene counts we observed.
- **Likelihood:** how well a possible answer explains the data.
- **Prior:** what the model thinks is reasonable before looking at this particular spot's data.
- **Posterior:** the updated beliefs after seeing the data.

Big idea

Bayesian inference is not just one best answer. It is a way to describe uncertainty. Instead of only saying “there are 4 B cells,” it can say “the abundance is probably around 4, but values around 3 or 5 are also possible.”

3.1 A non-biological analogy: guessing candies in a jar

Suppose you cannot see inside a jar, but you can hear a few clues when the jar shakes. You want to guess how many red, blue, and green candies are inside.

- The hidden numbers of candies are the **unknowns**.
- The sounds you hear are the **data**.
- Your belief that the jar probably has 20-50 candies, not 10,000, is a **prior**.
- A physical model connecting candies to sounds is the **likelihood**.
- Your final best guesses after hearing the sounds are the **posterior**.

Cell2location is similar, except the “jar” is a tissue spot, the “candy colors” are cell types, and the “sounds” are RNA counts from genes.

4 Which things have a model?

This is one of the most important points.

Common confusion

Students often ask: “Which one should have a model? The data? The parameters? The prior?” The answer is: a Bayesian model has a probability model for the observed data *given* unknown quantities, and probability models called priors for unknown quantities. You do not put a prior on the observed numbers after you have observed them; they are fixed data. You put priors on unknowns.

For cell2location, a helpful classification is:

Thing	Role	Explanation
Observed spatial count $d_{s,g}$	Data	The measured RNA count of gene g at spot s . This is what the model tries to explain.
Reference signature $g_{f,g}$	Input to mapping step	Typical expression of gene g for cell type f . Page 12 treats this as an input when mapping cell types to spots. It is usually estimated from single-cell data first.
Cell abundance $w_{s,f}$	Main unknown	How much cell type f is present at spot s . This is the main answer we want.
Expected count $\mu_{s,g}$	Hidden expected value	The model’s predicted average count for gene g at spot s .
Technical factors $m_g, s_{e,g}, y_s$	Unknown adjustment parameters	These help explain technology differences, background RNA, and spot-by-spot RNA capture.
Overdispersion $\alpha_{e,g}$	Noise parameter	Controls how noisy the count distribution is for gene g in experiment e .
Hyperparameters $\hat{N}, \alpha^y$	User-set prior knobs	These control priors: expected cells per spot and how much spot sensitivity can vary.

Big idea

The main unknown is $w_{s,f}$: the abundance of cell type f at spot s . Most of the model exists to estimate $w_{s,f}$ while not being fooled by noisy counts and technical effects.

5 Bayes’ rule for the actual cell2location formula

This section answers a very common question: in the general formula

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta)p(\theta)}{p(\text{data})},$$

what is θ for cell2location?

Big idea

Your instinct is right: the most important part of θ is $w_{s,f}$, the abundance of cell type f at spot s . But the full model has other unknown parameters too, such as m_g , y_s , $s_{e,g}$, and $\alpha_{e,g}$. These are not the main biological answer, but the model needs them so that technical effects do not get mistaken for cell-type abundance.

5.1 First split the unknowns into two groups

It is helpful to separate the unknowns into the main unknowns and the helper unknowns.

Model piece

$$\underbrace{W = \{w_{s,f}\}}_{\text{main biological unknowns}} \quad \underbrace{\Phi = \{m_g, y_s, s_{e,g}, \alpha_{e,g}\}}_{\text{helper / technical unknowns}}.$$

Here W is a big table. One row is a spatial spot, and one column is a cell type. For example, $w_{17,\text{astrocyte}}$ means “estimated astrocyte abundance in spot 17.”

The symbol Φ is just a convenient name for all the other unknowns. These other unknowns are sometimes called **nuisance parameters**. That name does *not* mean they are useless. It means they are not the final scientific answer we mainly care about, but we must still estimate or account for them.

5.2 The full Bayes’ rule version

Let

- $D = \{d_{s,g}\}$ be the observed spatial gene-count data.
- $\Gamma = \{g_{f,g}\}$ be the reference cell-type signatures from single-cell data.
- H be the hyperparameters, such as $\hat{N}$, α^y , and the chosen number R of shared tissue patterns.

Then the full Bayesian formula can be written as:

Model piece

$$p(W, \Phi \mid D, \Gamma, H) = \frac{\overbrace{p(D \mid W, \Phi, \Gamma)}^{\text{likelihood}} \overbrace{p(W, \Phi \mid H)}^{\text{prior}}}{\underbrace{p(D \mid \Gamma, H)}_{\text{evidence / normalizing denominator}}}$$

This is the same Bayes’ rule as before, but with names that match cell2location.

5.3 What is on the numerator?

The numerator is the important part to understand:

Model piece

$$\begin{aligned}\text{numerator} &= \text{likelihood} \times \text{prior} \\ &= p(D \mid W, \Phi, \Gamma) \times p(W, \Phi \mid H).\end{aligned}$$

The likelihood asks: “How well does this proposed explanation predict the observed counts?”
The prior asks: “How reasonable was this proposed explanation before seeing these counts?”

Now plug in the negative binomial observation model:

Model piece

$$p(D \mid W, \Phi, \Gamma) = \prod_{s=1}^S \prod_{g=1}^G \text{NB}(d_{s,g} \mid \mu_{s,g}, \alpha_{e(s),g})$$

$$\mu_{s,g} = \left(m_g \sum_{f=1}^F w_{s,f} g_{f,g} + s_{e(s),g} \right) y_s.$$

The notation $e(s)$ means “the experiment or batch that spot s belongs to.” If there is only one experiment, you can ignore this detail.

The prior part can be shown conceptually like this:

Model piece

$$p(W, \Phi \mid H) = \underbrace{p(W \mid \hat{N}, R, \dots)}_{\text{prior for cell abundances}} \underbrace{p(Y \mid \alpha^y, \dots)}_{\text{prior for spot sensitivities}} \underbrace{p(M)p(B)p(A)}_{\text{priors for other technical/noise parameters}}.$$

In that last line, $Y = \{y_s\}$, $M = \{m_g\}$, $B = \{s_{e,g}\}$, and $A = \{\alpha_{e,g}\}$. The page-12 Methods text does not list every prior distribution in the main article; it points to the Supplementary Methods for the complete derivation. For a beginner, the key idea is that the numerator is always:

$$\text{likelihood} \times \text{prior}.$$

5.4 What is the denominator doing?

The denominator is

Model piece

$$p(D \mid \Gamma, H) = \int p(D \mid W, \Phi, \Gamma) p(W, \Phi \mid H) dW d\Phi.$$

This means: add up the numerator over all possible values of the unknowns. Its job is to make the posterior probabilities add up to 1. In many huge models, this integral is too hard to calculate exactly, which is why cell2location uses approximate variational inference.

Common confusion

The denominator is not a new biological assumption. It is a normalizing constant. In practice, many explanations of Bayes' rule use $\propto$, meaning "proportional to," because the numerator is what compares one possible explanation against another.

5.5 If we only care about $w_{s,f}$

Usually, the final answer we want is $W = \{w_{s,f}\}$: the cell-type abundance map. But because m_g , y_s , $s_{e,g}$, and $\alpha_{e,g}$ are also unknown, a careful Bayesian answer averages over them:

Model piece

$$p(W \mid D, \Gamma, H) = \int p(W, \Phi \mid D, \Gamma, H) d\Phi.$$

In words:

To learn the cell abundances W , consider all plausible technical settings Φ , and average over their uncertainty.

So if a previous simple tutorial wrote θ , you can read it in two ways:

- **Simple version:** $\theta = W$, the cell abundance table.
- **Full version:** $\theta = (W, \Phi)$, all unknown quantities in the model.

5.6 A tiny numerical example

Suppose one spot has two possible cell types: neurons and astrocytes. For one gene, the reference signatures are:

$$g_{\text{neuron},g} = 2, \quad g_{\text{astrocyte},g} = 20.$$

Now imagine the model is testing this possible explanation:

$$w_{s,\text{neuron}} = 3, \quad w_{s,\text{astrocyte}} = 4, \quad m_g = 0.5, \quad s_{e,g} = 1, \quad y_s = 1.2.$$

Then the expected count is

$$\begin{aligned}
\mu_{s,g} &= \left(m_g \sum_f w_{s,f} g_{f,g} + s_{e,g} \right) y_s \\
&= (0.5(3 \cdot 2 + 4 \cdot 20) + 1) 1.2 \\
&= (0.5(86) + 1) 1.2 \\
&= 52.8.
\end{aligned}$$

If the observed count was $d_{s,g} = 50$, this proposed explanation would get a fairly good likelihood for this gene because 50 is close to 52.8. If the observed count was $d_{s,g} = 5$, this proposed explanation would get a much worse likelihood for this gene.

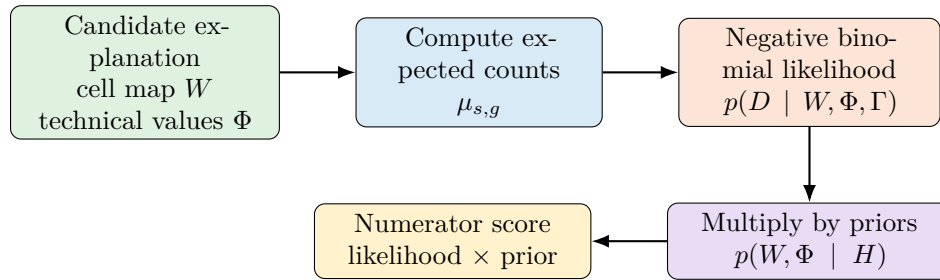


Figure 2: The numerator of Bayes’ rule gives a score to each possible explanation. Good explanations both predict the observed counts well and satisfy the prior assumptions.

5.7 Parameter or hyperparameter?

Here is the classification in the page-12 formula.

Symbol	Plain-English meaning	Role	Learned or chosen?
$d_{s,g}$	Observed count of gene g at spot s	Data	Observed, not learned
$g_{f,g}$	Reference expression of gene g in cell type f	Input to spatial model	Estimated earlier from single-cell data, then treated as fixed input for mapping
$w_{s,f}$	Abundance of cell type f at spot s	Main parameter	Learned from the spatial data
$\mu_{s,g}$	Expected count for gene g at spot s	Deterministic hidden mean	Computed from the parameters, not usually a separate free parameter
m_g	Gene-specific technology sensitivity multiplier	Parameter	Learned; helps adjust differences between technologies
$s_{e,g}$	Gene- and experiment-specific additive background shift	Parameter	Learned; helps account for contaminating/background RNA

y_s	Spot-specific RNA detection sensitivity	Parameter	Learned; its prior is controlled by α^y
$\alpha_{e,g}$	Gene- and experiment-specific overdispersion/noise	Parameter	Learned or estimated as part of the model's noise behavior
$\hat{N}$	Expected number of cells per spot	Hyperparameter	Chosen by the user, often using histology images
α^y	Strength of regularization for y_s	Hyperparameter	Chosen by the user; large values keep y_s closer to the batch average
R	Number of shared abundance patterns in the hierarchical prior	Model setting	Chosen before fitting; the paper usually sets $R = 50$

Check your understanding

A quick test: is y_s a hyperparameter because it has a prior? No. y_s is a parameter. The hyperparameter is α^y , because α^y controls the prior over y_s .

6 The page-12 model, rewritten gently

Page 12 describes the cell2location model in compact mathematical notation. Here is the main model in a beginner-friendly form.

6.1 Step 1: the observed counts are noisy counts

Model piece

$$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g})$$

This says:

The observed count $d_{s,g}$ for spot s and gene g is treated as a random count drawn from a negative binomial distribution with average level $\mu_{s,g}$ and noise/overdispersion $\alpha_{e,g}$.

Breakdown:

- s means a spatial spot or location.
- g means a gene.
- e means the experiment or batch containing that spot.
- $d_{s,g}$ is observed. Example: spot 17 has 42 counts of gene ACTB.
- $\mu_{s,g}$ is the model's expected average count.
- $\alpha_{e,g}$ lets the count be more variable than a simple Poisson count.

6.2 Why negative binomial?

Counts are whole numbers. A simple count model is the Poisson distribution, but real RNA counts are usually more variable than Poisson. The negative binomial distribution is a common way to model count data with extra variability.

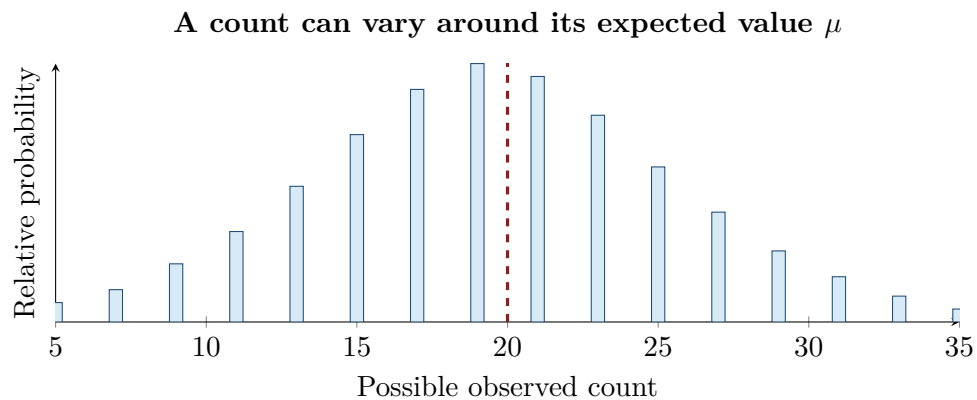


Figure 3: A schematic count distribution. Even if the expected count is about 20, the observed count might be 17, 22, or another nearby value. The negative binomial distribution allows more spread than the simplest count model.

Check your understanding

If a model predicts $\mu_{s,g} = 20$, does it require the observed count to be exactly 20? No. It says counts near 20 are more plausible than counts very far away, but random variation is allowed.

6.3 Step 2: expected gene counts are built from cell-type mixtures

The model defines $\mu_{s,g}$ like this:

Model piece

$$\mu_{s,g} = \left(\underbrace{m_g}_{\substack{\text{gene/technology} \\ \text{sensitivity}}} \underbrace{\sum_{f=1}^F w_{s,f} g_{f,g}}_{\substack{\text{cell-type} \\ \text{contributions}}} + \underbrace{s_{e,g}}_{\substack{\text{background} \\ \text{RNA shift}}} \right) \underbrace{y_s}_{\substack{\text{spot-level} \\ \text{sensitivity}}}$$

This formula is the heart of the model. It says:

The expected RNA count for a gene at a spot equals the sum of contributions from all possible cell types, corrected for technology, background RNA, and the spot's RNA-capture sensitivity.

6.4 The most important part: the sum over cell types

Focus first on this part:

$$\sum_{f=1}^F w_{s,f} g_{f,g}.$$

Here:

- F is the number of reference cell types.
- f is one cell type, such as T cell, B cell, astrocyte subtype, neuron subtype, and so on.
- $g_{f,g}$ is the reference signature: how much gene g is expected in one unit of cell type f .
- $w_{s,f}$ is the estimated amount of cell type f at spot s .

Big idea

The sum is a weighted recipe. Each cell type has a gene-expression recipe. The weights $w_{s,f}$ say how much of each recipe to mix into spot s .

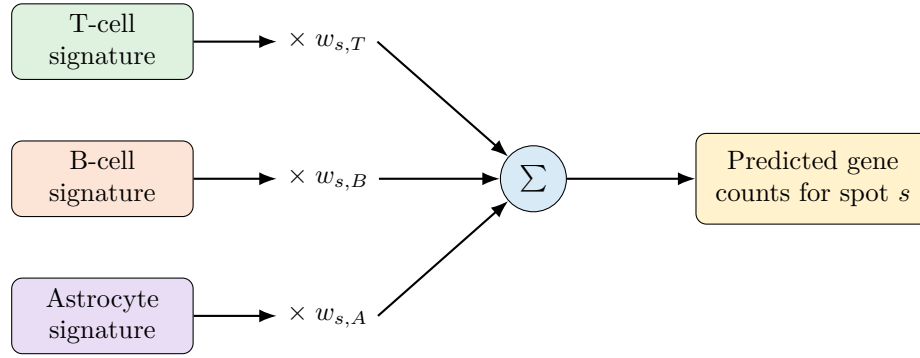


Figure 4: Each reference signature is multiplied by its abundance in the spot, then the contributions are added. This is why the model is called a decomposition model: it decomposes a mixed spot into cell-type pieces.

7 A worked example with small numbers

Let us use only two cell types and three genes. This is much smaller than real data, but it shows the idea.

7.1 Reference signatures

Suppose single-cell data tell us the typical gene counts per cell type look like this:

Reference signature	Gene A	Gene B	Gene C
T cell	8	0	4
B cell	0	7	3

Interpretation:

- Gene A is high in T cells and low in B cells.
- Gene B is high in B cells and low in T cells.
- Gene C is expressed by both.

Now suppose one spatial spot has observed counts:

Observed spot	Gene A	Gene B	Gene C
Spot s	15	25	16

7.2 Try a possible answer

Maybe spot s contains about 2 T cells and 3 B cells. That means:

$$w_{s,T} = 2, \quad w_{s,B} = 3.$$

Then the expected contributions are:

Contribution	Gene A	Gene B	Gene C
2 T cells	$2 \cdot 8 = 16$	$2 \cdot 0 = 0$	$2 \cdot 4 = 8$
3 B cells	$3 \cdot 0 = 0$	$3 \cdot 7 = 21$	$3 \cdot 3 = 9$
Predicted total μ	16	21	17
Observed count d	15	25	16

This is not perfect, but it is close. Because the count model is noisy, close can still be very likely.

7.3 Try a bad answer

If the spot had 0 T cells and 5 B cells, then Gene A would be predicted near 0, but the observed Gene A count is 15. So that answer is less plausible.

Candidate mixture	Predicted Gene A	Predicted Gene B	Good explanation?
2 T cells, 3 B cells	16	21	Reasonably good
0 T cells, 5 B cells	0	35	Bad for Gene A
5 T cells, 0 B cells	40	0	Bad for Gene B

Big idea

The likelihood gives higher probability to mixtures whose predicted counts are close to the observed counts, while still allowing noise.

8 Now add the technical correction terms

The simple mixture $\sum_f w_{s,f} g_{f,g}$ is not enough for real data. Page 12 adds three correction terms:

$$\mu_{s,g} = \left(m_g \sum_f w_{s,f} g_{f,g} + s_{e,g} \right) y_s.$$

8.1 The gene/technology sensitivity term m_g

m_g adjusts for gene-specific differences between technologies or datasets. Maybe the reference signatures came from one type of RNA-seq, but the spatial data came from another. Some genes may appear systematically higher or lower in one technology.

Symbol	Meaning
m_g	A gene-specific multiplier. It says whether gene g tends to be detected more or less strongly in the spatial technology compared with the reference.

If $m_g = 2$, then the model doubles the contribution of gene g before comparing to spatial counts. If $m_g = 0.5$, it halves it.

8.2 The additive background term $s_{e,g}$

$s_{e,g}$ allows for extra RNA not explained by cells in the spot. In experiments, a little RNA can float around or contaminate a region. This background can depend on the experiment e and gene g .

Symbol	Meaning
$s_{e,g}$	A gene- and experiment-specific additive shift. It can explain background or contaminating RNA.

Important: $s_{e,g}$ is *added*. The model says, roughly, “Even before considering spot-specific capture sensitivity, maybe this experiment has some extra background for this gene.”

8.3 The spot sensitivity term y_s

y_s is a multiplier for a whole spot. Some spots capture more RNA than others because of tissue quality, chemistry, or technical differences.

Symbol	Meaning
y_s	A spot-specific multiplier. If spot s captures twice as much RNA, y_s might be larger; if it captures less RNA, y_s might be smaller.

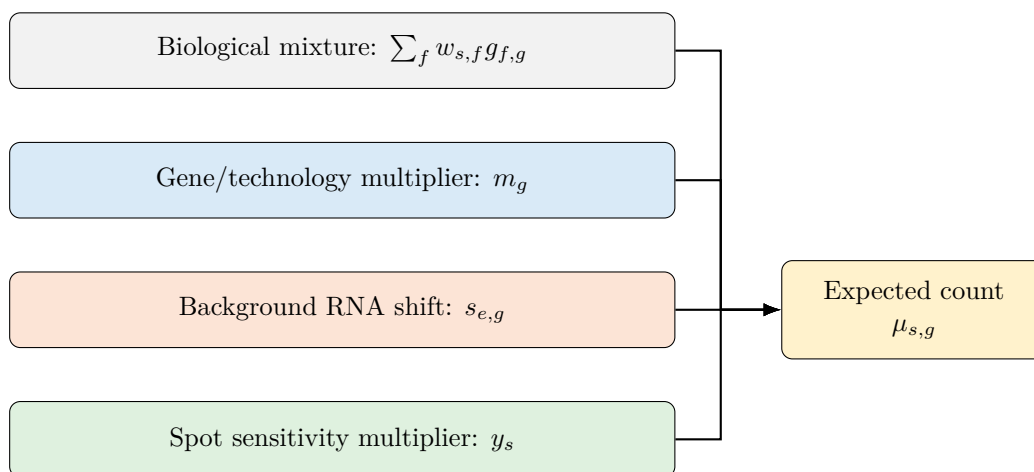


Figure 5: The expected count $\mu_{s,g}$ is not only biology. It also includes technical adjustments.

Common confusion

Without y_s , a spot with low RNA capture might look like it has fewer cells. Without m_g , technology differences can look like biological differences. Without $s_{e,g}$, background RNA can be mistaken for a cell type. These correction terms help the model avoid those mistakes.

9 What exactly is $w_{s,f}$?

The symbol $w_{s,f}$ is the star of the model.

Model piece

$w_{s,f}$ = estimated abundance of cell type f at spot s .

If $w_{s,B} = 3.2$, the model is saying that spot s contains an amount of B-cell RNA roughly comparable to 3.2 B cells. This number can be fractional because a spot may contain partial cells, because reference signatures are averages, and because the model estimates abundance rather than literally counting individual cell bodies.

9.1 Absolute abundance versus proportion

There are two related ideas:

- **Absolute abundance:** about how many cells of a type are in the spot. Example: 3 B cells and 2 T cells.
- **Proportion:** what fraction of the spot's cells are each type. Example: B cells are $3/(3+2) = 60\%$.

Cell2location aims to estimate absolute abundance, and it uses a prior about the expected number of cells per spot to help do this.

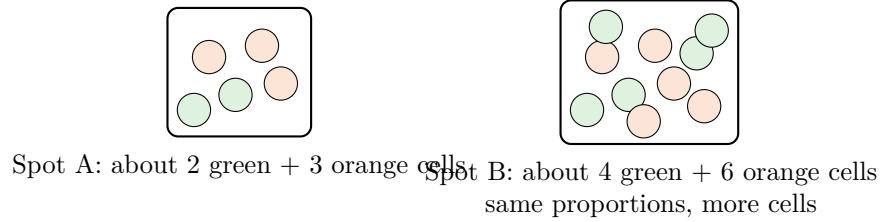


Figure 6: Two spots can have the same proportions but different absolute cell abundances. Cell2location tries to estimate abundance, not only percentages.

10 The model as a generative story

A **generative story** describes how data could be produced if we knew all the hidden quantities. Cell2location's story can be simplified as follows.

1. For each spot s , choose hidden cell-type abundances $w_{s,1}, \dots, w_{s,F}$.
2. Use reference signatures $g_{f,g}$ to predict how much each gene should appear from that mixture.
3. Adjust for technology with m_g , background RNA with $s_{e,g}$, and spot sensitivity with y_s .
4. This gives an expected count $\mu_{s,g}$ for every spot and gene.
5. Draw the observed count $d_{s,g}$ from a negative binomial distribution centered around $\mu_{s,g}$.

Bayesian inference reverses this. We already have the observed $d_{s,g}$. We want to infer the likely $w_{s,f}$ and other unknowns.

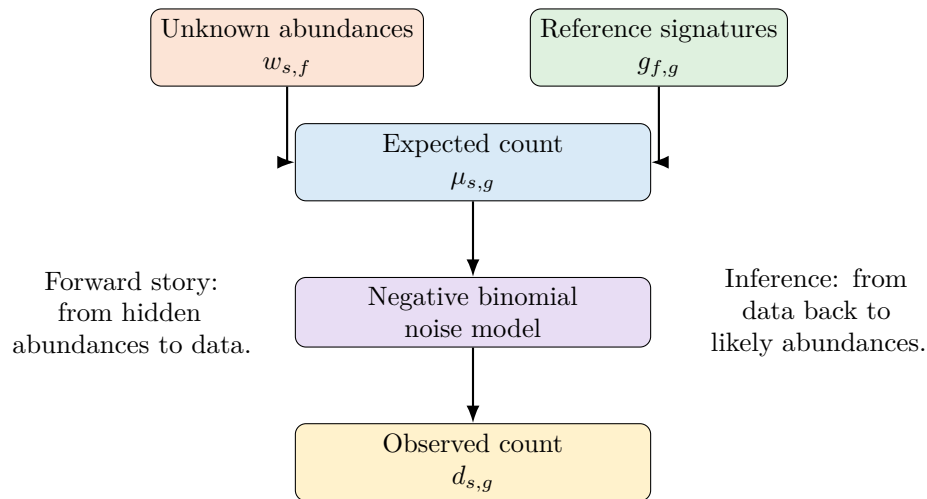


Figure 7: The generative direction goes from hidden cell abundance to observed counts. Inference goes in the reverse direction.

11 Where do priors enter?

A prior is a probability model for an unknown before the current data are used. It is not random guessing. It is a way to include reasonable background information.

In cell2location, priors help with questions such as:

- Are cell abundances allowed to be negative? No. Priors keep $w_{s,f} \geq 0$.
- Is it reasonable to estimate 400 cells in one Visium spot? Usually no. The expected cell abundance prior discourages unreasonable totals.
- Should neighboring or similar tissue locations have completely unrelated cell mixtures? Usually no. The hierarchical prior encourages sharing information.
- Should spot sensitivity y_s vary wildly from spot to spot? Usually no, unless the data suggest strong technical gradients.

Big idea

The likelihood asks, “Which values explain the counts?” The prior asks, “Which values were reasonable before seeing these counts?” The posterior balances both.

11.1 Priors do not erase data

A prior is not a command. If the data strongly disagree with the prior, the posterior can move away from the prior. But if data are weak or ambiguous, the prior has more influence.

This is especially important for rare or fine-grained cell types. If a rare cell type has only a weak signal, the hierarchical prior can help by borrowing information from other spots that look similar.

12 What are hyperparameters?

A parameter is an unknown inside the model. A hyperparameter is a setting that controls a prior or model behavior.

Word	Meaning
Parameter	A quantity the model estimates from the data, such as $w_{s,f}$ or y_s .
Prior	A probability distribution placed on an unknown parameter.
Hyperparameter	A knob that controls the prior. Often set by the user using outside knowledge.
Posterior	The updated distribution over parameters after seeing the data.

12.1 Analogy: a teacher grading essays

Suppose a teacher expects high-school essays to be around 500 words. This expectation is not the essay itself. It is a guideline.

- The actual essay length is like a **parameter**.
- The belief that essays are usually around 500 words is like a **prior**.
- The number 500 is like a **hyperparameter**.

In cell2location, $\hat{N}$ is like saying, “I expect about this many cells per spot.” The model can adjust based on data, but this expectation guides absolute cell abundance estimates.

13 The two user-adjusted hyperparameters on page 12

Page 12 highlights two parameters that users should adjust based on the dataset.

Model piece

$\hat{N}$ = expected cell abundance per spot

α^y = regularization strength for spot sensitivity y_s

13.1 Hyperparameter 1: expected cell abundance per spot

$\hat{N}$ tells the model roughly how many cells are expected in a spot. It can be estimated from histology images by counting nuclei in 10-20 locations and taking the average.

Big idea

$\hat{N}$ is especially important for estimating *absolute* cell abundance. If you want the model to say “about 8 cells” rather than only “about 40 percent T cells,” you need a reasonable idea of how many cells a spot usually contains.

13.1.1 Example

If a tissue image shows about 8 nuclei per spot, you might set $\hat{N} \approx 8$.

If you set $\hat{N}$ much too high, the model may be nudged toward too many cells per spot. If you set it much too low, it may be nudged toward too few. The paper notes the model is robust to a range of similar values, but good choices help.

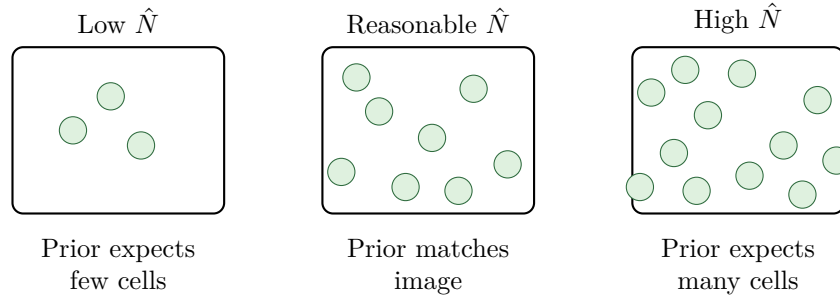


Figure 8: $\hat{N}$ is a prior expectation about total cells per spot. It should be estimated from the tissue image when possible.

13.2 Hyperparameter 2: regularization of spot sensitivity

α^y controls the prior on y_s , the spot-level RNA detection sensitivity.

Page 12 says α^y is the shape parameter of a gamma prior over y_s . For intuition, you do not need the full gamma math. Think of α^y as controlling how tightly the model expects y_s to stay near the experiment's average sensitivity.

Choice of α^y	Intuition
Large α^y, such as 200	Strong regularization. Spot sensitivities y_s are expected to be close to the batch average. Use when RNA detection is fairly uniform across spots.
Smaller α^y, such as 20	Weaker regularization. Spot sensitivities y_s are allowed to vary more. Use when there are strong spatial gradients or uneven RNA detection.

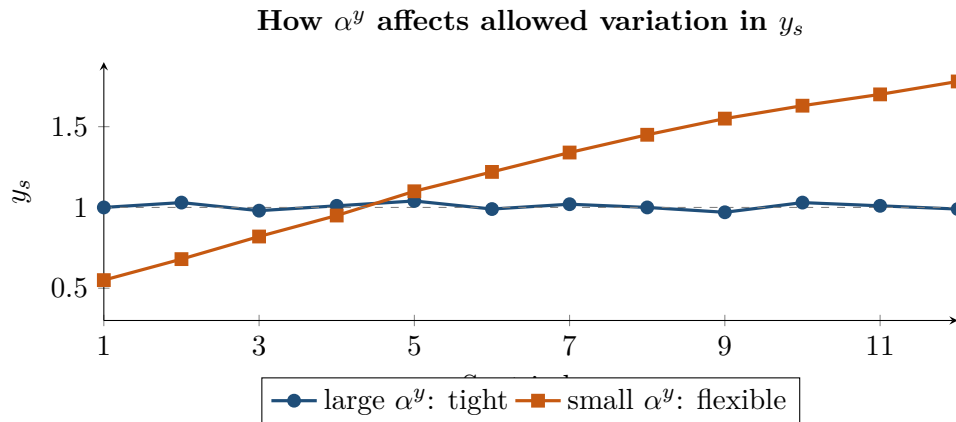


Figure 9: Schematic only. A large α^y keeps spot sensitivity near the average. A smaller α^y allows stronger spot-to-spot sensitivity differences.

Common confusion

Regularization means “do not let this parameter wiggle too much unless the data really need it.” Large α^y means stronger regularization for y_s .

14 Which prior distribution does each parameter have?

You asked an important question: “For the parameters, should they follow some distribution controlled by hyperparameters?”

The Bayesian answer is: usually, yes. In a Bayesian model, an unknown parameter is usually given a **prior distribution**. That prior distribution is often controlled by one or more **hyperparameters**. However, in a large model like cell2location, not every symbol in the equation is the same kind of object.

Big idea

A simple Bayesian model might look like

parameter $\sim$ prior controlled by hyperparameters,

then

data $\sim$ likelihood controlled by the parameter.

Cell2location follows this idea, but with several layers of parameters. Some parameters have priors directly; others are built from hidden intermediate variables that also have priors.

For this section, I will use the **Gamma(shape, rate)** convention:

$$X \sim \text{Gamma}(a, b) \implies \mathbb{E}[X] = \frac{a}{b}.$$

The first number is the **shape**; the second number is the **rate**. This matters because some books use “scale” instead of “rate.”

Common confusion

The paper’s page-12 Methods page gives the main count model and names the two user-adjusted hyperparameters, $\hat{N}$ and α^y . The full implementation has more internal prior constants than the paper highlights. So, in this section, think of the detailed formulas as a beginner-friendly map of the model’s prior structure, not as the only way one could write a Bayesian spatial model.

14.1 A first map: data, parameters, inputs, and hyperparameters

The page-12 model is

$$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g})$$

with

$$\mu_{s,g} = \left(m_g \sum_f w_{s,f} g_{f,g} + s_{e,g} \right) y_s.$$

A useful classification is:

Symbol	Role	Plain-English meaning
$d_{s,g}$	observed data	the count already measured for gene g in spot s
$w_{s,f}$	parameter	the cell-type abundance we mainly want to estimate
m_g	parameter	gene-specific technology correction
y_s	parameter	spot-specific RNA detection sensitivity
$s_{e,g}$	parameter	background / contaminating RNA term
$\alpha_{e,g}$	parameter	negative-binomial overdispersion, or extra count noise
$g_{f,g}$	input in this spatial step	reference expression of gene g in cell type f
$\hat{N}, \alpha^y$	hyperparameters	user-adjusted settings that control priors

So your intuition was partly right: if we write a generic Bayesian formula such as

$$p(\theta \mid D),$$

then $w_{s,f}$ is the most important part of θ . But in the full model,

$$\theta = \{w_{s,f}, m_g, y_s, s_{e,g}, \alpha_{e,g}, \text{extra hidden prior variables}\}.$$

14.2 Visual ladder: hyperparameter to prior to parameter to data

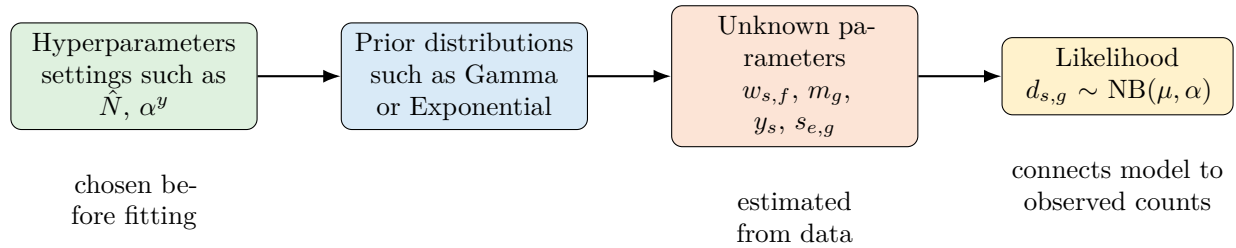


Figure 10: The role of hyperparameters, priors, parameters, and data in a Bayesian model.

14.3 Observed data: $d_{s,g}$

This is **not** a parameter. This is the measured data.

$$d_{s,g} = \text{observed count of gene } g \text{ at spatial location } s.$$

The model assumes

$$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g}).$$

In plain English:

Once the model chooses possible values for all unknown parameters, it predicts an average count $\mu_{s,g}$. The actual observed count $d_{s,g}$ is then allowed to wiggle around that average using a negative binomial distribution.

This distribution is called the **likelihood**. It is the probability model for the data after the unknown parameters have been chosen.

14.4 Cell abundance: $w_{s,f}$

This is the most important parameter for biological interpretation.

$w_{s,f}$ = abundance of cell type f at location s .

This is the closest thing to the “answer” that cell2location wants. It asks: how much of each cell type is present at each spot?

But $w_{s,f}$ does not get only a one-line prior such as

$$w_{s,f} \sim \text{Gamma}(\dots).$$

Instead, it receives a **hierarchical factorization prior**. That means the prior is built in several steps.

A simplified version is:

$$n_s \sim \text{Gamma}(\hat{N}\rho_N, \rho_N),$$

where n_s represents the total expected number of cells at location s and ρ_N is a prior-strength constant. Because

$$\mathbb{E}[n_s] = \frac{\hat{N}\rho_N}{\rho_N} = \hat{N},$$

the hyperparameter $\hat{N}$ says:

Before seeing the RNA data, we expect about $\hat{N}$ cells in a typical spot.

Then the model creates hidden tissue-zone variables. A beginner-friendly way to write the idea is:

$$\bar{w}_{s,f} = \sum_r z_{s,r} x_{r,f}.$$

Here:

- $z_{s,r}$ says how much tissue zone or compartment r is present at spot s .
- $x_{r,f}$ says how strongly cell type f belongs to tissue zone r .
- $\bar{w}_{s,f}$ is the prior mean for cell type f at spot s .

Finally, the abundance is modeled as something like

$$w_{s,f} \sim \text{Gamma}(\rho_w \bar{w}_{s,f}, \rho_w),$$

so that

$$\mathbb{E}[w_{s,f}] = \bar{w}_{s,f}.$$

Big idea

The prior for $w_{s,f}$ is hierarchical because the model first imagines shared tissue zones, then uses those zones to guide cell-type abundance estimates. This lets similar spots borrow information from each other.

So, for $w_{s,f}$:

$w_{s,f}$ is a parameter,

while

$\hat{N}, R, \rho_N, \rho_w$ and related prior constants are hyperparameters.

The main user-facing hyperparameter here is $\hat{N}$.

14.5 Gene-specific technology scaling: m_g

m_g = gene-specific scaling factor.

This is a **parameter**, not a hyperparameter.

Its job is to correct for this problem:

The same gene might be detected differently in the single-cell reference data and in the spatial data.

For example, gene g might look strong in single-cell RNA-seq but weaker in Visium spatial data. Then m_g adjusts for that.

A simplified version of the prior is:

$$m_g \sim \text{Gamma} \left(\alpha_m, \frac{\alpha_m}{m_{\text{mean}}} \right).$$

Because we use the shape-rate convention,

$$\mathbb{E}[m_g] = m_{\text{mean}}.$$

The model also places priors on the mean and variation-controlling variables, such as

$$m_{\text{mean}} \sim \text{Gamma}(\text{hyperparameter values})$$

and an Exponential prior on a transformed variable that controls α_m .

Plain English:

m_g is like a correction knob for each gene. The prior says the knob should usually be near a reasonable value, unless the data strongly suggest otherwise.

So:

m_g is a parameter.

The values controlling its Gamma prior are hyperparameters.

14.6 Location sensitivity: y_s

y_s = RNA detection sensitivity at location s .

This is also a **parameter**, not a hyperparameter.

Its job is to correct for the fact that some spatial spots capture more RNA than others. Maybe two spots contain similar cell types, but one spot has stronger sequencing signal. Then y_s helps adjust for that.

The prior is

$$y_s \sim \text{Gamma}\left(\alpha^y, \frac{\alpha^y}{y_e}\right),$$

where y_e is the average detection sensitivity for experiment or batch e . The batch-level average also has a prior, often written as

$$y_e \sim \text{Gamma}\left(10, \frac{10}{\text{detection mean}}\right).$$

Because

$$\mathbb{E}[y_s \mid y_e] = y_e,$$

the prior says:

Each location's detection sensitivity should usually be close to the experiment average y_e .

The hyperparameter α^y controls how strict this is.

α^y value	Meaning
Large, such as 200	Strong regularization: y_s stays close to the batch average y_e .
Smaller, such as 20	Weaker regularization: y_s can vary more across spots.

So:

y_s is a parameter, α^y is a hyperparameter.

14.7 Additive background or contamination: $s_{e,g}$

$s_{e,g}$ = extra expression of gene g in experiment e

that cannot be explained by the cell-type mixture.

This is a **parameter**, not a hyperparameter.

Its job is to handle background RNA, sometimes called free-floating or contaminating RNA.

A simplified Gamma-based prior is:

$$s_{e,g} \sim \text{Gamma}\left(\alpha_e^s, \frac{\alpha_e^s}{s_e^{\text{mean}}}\right).$$

So

$$\mathbb{E}[s_{e,g}] = s_e^{\text{mean}}.$$

The model also puts priors on background mean and variation-controlling variables, for example:

$$s_e^{\text{mean}} \sim \text{Gamma}(a_s, b_s),$$

with additional Gamma or Exponential hierarchy for the variation.

Plain English:

$s_{e,g}$ is the model's background-noise correction. It says: maybe some reads for gene g appear even if the cell-type mixture does not fully explain them.

So:

$$s_{e,g} \text{ is a parameter.}$$

The constants controlling its prior are hyperparameters.

14.8 Overdispersion: $\alpha_{e,g}$

$\alpha_{e,g}$ = overdispersion parameter for gene g in experiment e .

This is a **parameter**. It controls how noisy the counts are around the mean $\mu_{s,g}$.

One common way to think about the negative binomial is:

$$\text{Var}(d_{s,g}) = \mu_{s,g} + \frac{\mu_{s,g}^2}{\alpha_{e,g}}.$$

So smaller $\alpha_{e,g}$ means more extra noise.

The implementation uses a transformed version. Let

$$q_{e,g} = \alpha_{e,g}^{-1/2}.$$

Then the prior can be written as

$$q_{e,g} \sim \text{Exponential}(\phi_\alpha),$$

with

$$\phi_\alpha \sim \text{Gamma}(9, 3),$$

and then

$$\alpha_{e,g} = \frac{1}{q_{e,g}^2}.$$

Plain English:

$\alpha_{e,g}$ tells the model how much random count noise to expect for each gene and experiment.

So:

$$\alpha_{e,g} \text{ is a parameter,}$$

while

ϕ_α and the constants in its Gamma prior are hyperparameters.

14.9 Reference signatures: $g_{f,g}$

$g_{f,g}$ = expected expression of gene g in cell type f .

In the page-12 spatial mapping model, $g_{f,g}$ is usually treated as **given input**, not as an unknown parameter being estimated in that same step.

This can feel confusing because $g_{f,g}$ was estimated earlier from single-cell or single-nucleus RNA-seq. So there are two levels of the workflow:

Workflow step	Role of $g_{f,g}$
Reference-building step	$g_{f,g}$ is estimated from single-cell data, usually with a count model.
Spatial mapping step	$g_{f,g}$ is treated as known input and used to estimate $w_{s,f}$.

So, for the page-12 spatial model:

$g_{f,g}$ is treated as known input.

But in the full workflow:

$g_{f,g}$ was estimated earlier from another model.

14.10 Summary table: parameter or hyperparameter?

Symbol	Meaning	Role	Prior / distribution idea
$d_{s,g}$	observed gene count	data	$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g})$
$w_{s,f}$	cell-type abundance	parameter	hierarchical Gamma factorization prior
$\hat{N}$	expected cells per spot	hyperparameter	controls prior for total cells and absolute abundance scale
m_g	gene technology correction	parameter	Gamma prior, with hyperpriors on mean/variation
y_s	spot detection sensitivity	parameter	$y_s \sim \text{Gamma}(\alpha^y, \alpha^y/y_e)$
α^y	regularization for y_s	hyperparameter	controls how much y_s can vary
$s_{e,g}$	background RNA	parameter	Gamma prior, with hyperpriors on mean/variation
$\alpha_{e,g}$	NB overdispersion	parameter	indirect Exponential/Gamma prior through $\alpha_{e,g}^{-1/2}$
$g_{f,g}$	reference signature	input here	estimated earlier, then treated as fixed input

Big idea

The biggest idea is:

parameters are unknown quantities the model estimates,

while

hyperparameters are settings that control the priors on those unknowns.

For cell2location, the main biological parameter of interest is $w_{s,f}$. The other parameters, such as m_g , y_s , $s_{e,g}$, and $\alpha_{e,g}$, help the model avoid mistaking technical noise for real cell-type abundance.

15 The hierarchical part: borrowing strength across locations

Page 12 says the cell type abundance estimates $w_{s,f}$ are not independent across locations. Why? Because tissues are organized into regions. Nearby or similar spots often have similar mixtures of cell types.

For example, in brain tissue, one region may mostly contain one group of neuron subtypes, while another region contains a different group. In a lymph node, some regions contain B-cell zones and others contain T-cell zones.

Big idea

“Borrowing statistical strength” means using patterns learned from many related spots to help estimate a difficult spot. This is useful when the signal for a rare or fine-grained cell type is weak at one location.

15.1 A school analogy

Suppose one student misses part of a quiz, but you know they studied with a group and their answers on other questions are similar to that group. You would not copy the group’s answers blindly, but the group gives useful context.

Similarly, a spot with noisy data can be helped by other spots that appear to belong to the same tissue zone.

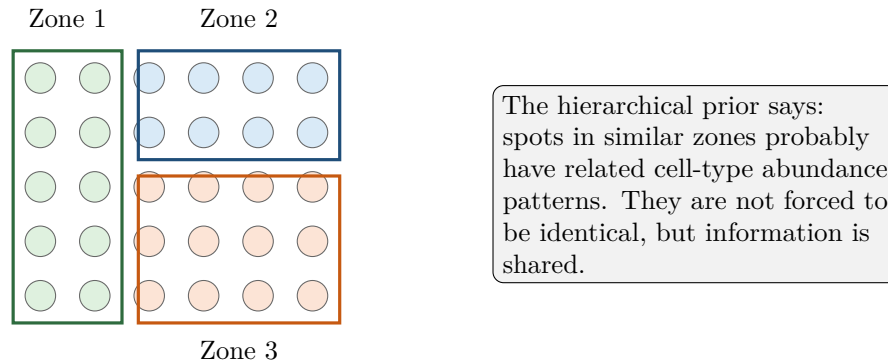


Figure 11: A schematic tissue grid. Spots in the same zone often share cell compositions. The hierarchical model uses that fact.

15.2 What is the “factorization” idea?

Page 12 says $w_{s,f}$ is modeled with a hierarchical decomposition prior, assuming R groups of cell types that partly share common abundance profiles. Unless stated otherwise, $R = 50$ in the paper.

A simplified way to imagine this is:

cell abundance at spot $s \approx$ combination of several tissue-zone patterns.

This does *not* mean there are exactly 50 biological tissue zones. Instead, $R = 50$ is a flexible model setting that gives the model enough reusable patterns to represent many spatial structures.

16 Reference cell-type signatures

Cell2location needs reference signatures $g_{f,g}$. These usually come from single-cell or single-nucleus RNA-seq data.

Model piece

$g_{f,g}$ = expected expression of gene g in cell type f .

16.1 Why signatures matter

If the reference says T cells express Gene A and B cells express Gene B, then a spot with lots of Gene A is evidence for T cells, and a spot with lots of Gene B is evidence for B cells.

But the reference must be reasonably comprehensive. If a cell type is missing from the reference, the model may try to explain its RNA using the closest available cell types.

Assumption

The model assumes the reference signatures are good enough to explain the spatial data. They do not have to be perfect, but important missing cell types or badly mismatched signatures can hurt the results.

16.2 Why use negative binomial regression to estimate signatures?

Page 12 says the signature matrix is often estimated using a negative binomial regression model. The reason is that single-cell data also have count noise and batch effects. A model-based signature estimate can combine data across batches and technologies more robustly than simply averaging counts.

However, the page also notes that if batch effects are small, a simpler per-cluster average can be enough. For some non-UMI technologies, the negative binomial count model may not be appropriate, so a simpler method can be recommended.

17 The full model in plain English

Here is the page-12 model translated sentence by sentence.

Mathematical phrase	Plain-English translation
$d_{s,g}$	The count we observed for gene g in spot s .
$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g})$	The observed count is noisy. It tends to be near $\mu_{s,g}$, but can vary. $\alpha_{e,g}$ controls extra count variability.
$\mu_{s,g}$	The expected count before random noise is added.
$\sum_f w_{s,f} g_{f,g}$	Add up contributions from every cell type. For each cell type: abundance in the spot times that cell type's expression of the gene.
m_g	Adjusts gene g for technology-specific sensitivity.
$s_{e,g}$	Adds possible background or contaminating RNA for gene g in experiment e .
y_s	Adjusts for how much RNA was detected at spot s overall.
$w_{s,f}$	The main answer: abundance of cell type f in spot s .
Hierarchical prior on $w_{s,f}$	Spots in similar tissue regions share information, helping noisy or rare-cell estimates.
Variational inference	A fast approximate way to find the posterior distributions of the unknowns.

18 How Bayesian inference works here, step by step

18.1 Step A: propose possible cell mixtures

For a spot, the model considers many possible values of $w_{s,f}$. For example:

Candidate	T cells	B cells	Astrocytes
1	0	5	0
2	2	3	0
3	1	1	4
4	5	0	0

Real cell2location does not literally list only four candidates. It uses continuous probability distributions and optimization. But this mental picture is useful.

18.2 Step B: compute predicted gene counts

For each candidate, use the reference signatures and technical corrections to compute $\mu_{s,g}$ for every gene.

18.3 Step C: compare predictions to observed counts

The negative binomial likelihood gives a score: how likely are the observed counts if this candidate mixture were true?

18.4 Step D: combine with priors

Candidates that have impossible or unreasonable abundances are discouraged by priors. For example, negative cell abundance is impossible, and a total of 100 cells in a spot with $\hat{N} = 8$ is unlikely unless the data strongly demand it.

18.5 Step E: output posterior estimates

The final output is a posterior distribution for the cell abundances. In practice, software often reports summary values such as posterior mean abundance.

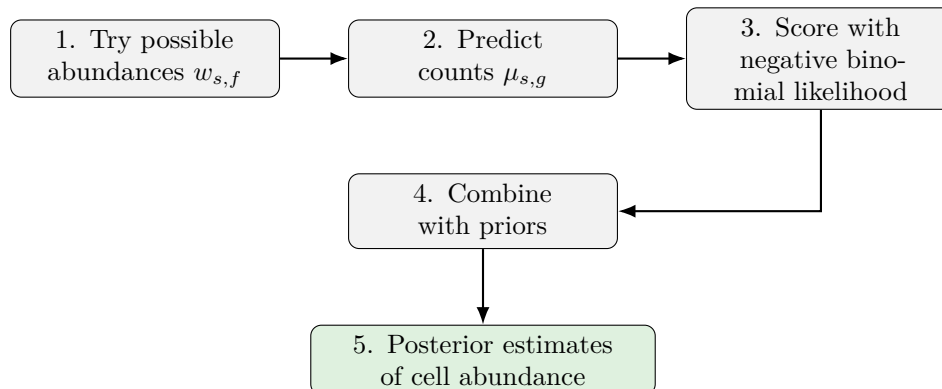


Figure 12: A beginner-level view of inference. The real algorithm is more advanced, but it follows this logic.

19 Variational inference: why the paper says “approximate”

A full Bayesian posterior can be very complicated. Cell2location has thousands or millions of unknowns: abundances for many spots and cell types, gene effects, batch effects, sensitivities, and noise parameters.

Instead of calculating the exact posterior, the software uses **approximate variational inference**. Here is the idea:

- Pick a flexible but manageable family of probability distributions.
- Adjust that distribution until it is close to the true posterior.
- Use optimization, often with GPU acceleration, to do this fast.

Big idea

Variational inference is like drawing a simpler map of a complicated city. The map is not the city itself, but if it is accurate enough, it helps you navigate quickly.

20 Assumptions made by the model

Every model has assumptions. Good science means knowing them.

Assumption

Assumption 1: additive mixtures. The gene expression in a spot is approximately a sum of contributions from cell types. This is the main mixture assumption.

Assumption

Assumption 2: reference signatures are useful. The signatures $g_{f,g}$ represent the cell types in the tissue well enough. Missing or incorrect cell types can mislead the model.

Assumption

Assumption 3: negative binomial count noise. Given $\mu_{s,g}$ and $\alpha_{e,g}$, the observed counts behave like negative binomial count data. This is common for RNA counts but is still a modeling choice.

Assumption

Assumption 4: technical effects can be summarized by model terms. Gene sensitivity m_g , background shift $s_{e,g}$, and spot sensitivity y_s are assumed to capture important technical variation.

Assumption

Assumption 5: locations can share information. The hierarchical prior assumes tissue organization creates repeated spatial patterns, so spots are not estimated in total isolation.

21 Common misunderstandings

21.1 Misunderstanding 1: “The model knows the true number of cells”

No. It estimates abundance from noisy data. The output is an informed estimate, not direct observation of every cell.

21.2 Misunderstanding 2: “A hyperparameter is another cell abundance”

No. A hyperparameter is a setting that shapes a prior. $\hat{N}$ is not the number of cells in one specific spot; it is a prior expectation for a typical spot.

21.3 Misunderstanding 3: “The prior is cheating”

No. Priors are part of Bayesian modeling. They make assumptions explicit. For example, knowing from an image that a spot usually has about 8 cells is useful information, not cheating.

21.4 Misunderstanding 4: “If the model has many terms, it must be arbitrary”

Not necessarily. Each term has a job:

Term	Job
$w_{s,f}$	biology: cell abundance
$g_{f,g}$	biology/reference: cell-type expression pattern
m_g	technology-specific gene scaling
$s_{e,g}$	background RNA
y_s	spot sensitivity
$\alpha_{e,g}$	extra count noise

21.5 Misunderstanding 5: “A high count for one gene proves one cell type is present”

Not always. One gene can be shared by multiple cell types, and counts are noisy. Cell2location uses many genes together, which is much stronger than relying on one marker gene.

22 A complete summary diagram

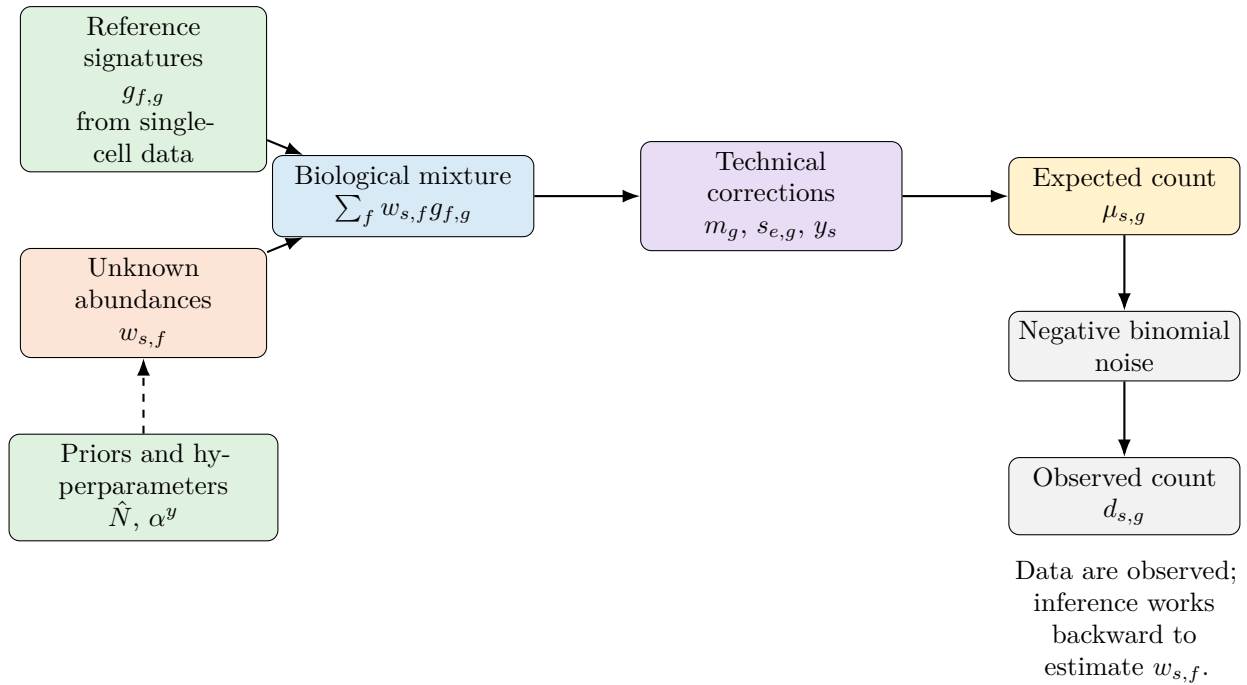


Figure 13: The cell2location model in one picture.

23 How to read the original page-12 equation

When you see this equation:

$$\mu_{s,g} = \left(m_g \sum_f w_{s,f} g_{f,g} + s_{e,g} \right) y_s,$$

read it slowly from the inside outward:

1. $g_{f,g}$: cell type f 's usual expression of gene g .
2. $w_{s,f} g_{f,g}$: contribution of cell type f to gene g in spot s .
3. $\sum_f w_{s,f} g_{f,g}$: total biological contribution from all cell types.
4. $m_g(\dots)$: adjust gene g for technology sensitivity.
5. $+s_{e,g}$: add background RNA for gene g in experiment e .
6. $\times y_s$: adjust the whole spot for RNA-capture sensitivity.
7. Result $\mu_{s,g}$: expected count of gene g in spot s .
8. Then $d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g})$: observed counts are noisy around that expectation.

Check your understanding

Try explaining this sentence out loud: “The observed count is noisy, so the model first predicts an expected count from a mixture of cell-type signatures, then uses a negative binomial distribution to describe the random count we observe.” If that sentence makes sense, you understand the main idea.

24 Mini practice questions

1. If a spot has high counts for genes that are usually high in B cells, what should happen to $w_{s,B}$?
2. If all genes in a spot are lower than expected, but the pattern across genes looks similar, which term might explain this: $w_{s,f}$, m_g , $s_{e,g}$, or y_s ?
3. If one gene is too high in every spot of one experiment, which term might help explain that?
4. Why is $\hat{N}$ important for absolute abundance?
5. Why might a hierarchical prior help with a rare cell type?

Answers

1. $w_{s,B}$ should become larger, if the pattern fits B cells better than other cell types.
2. y_s , because it changes the overall sensitivity of the spot.
3. $s_{e,g}$ or m_g , because those are gene- and experiment/technology-related terms.
4. Because it tells the model roughly how many cells are expected per spot, helping convert proportions into absolute cell counts.
5. Rare cell types have weak signals. Sharing information across similar spots can make the estimate more stable.

25 One-page cheat sheet

Cell2location page-12 model cheat sheet

Goal. Estimate $w_{s,f}$, the abundance of cell type f at spot s .

Observed data. $d_{s,g}$ = RNA count for gene g at spot s .

Reference input. $g_{f,g}$ = typical expression of gene g in cell type f .

Expected count.

$$\mu_{s,g} = \left(m_g \sum_f w_{s,f} g_{f,g} + s_{e,g} \right) y_s.$$

Count noise.

$$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g}).$$

Technical terms.

- m_g : gene/technology sensitivity.
- $s_{e,g}$: background or contaminating RNA.
- y_s : spot-level RNA detection sensitivity.
- $\alpha_{e,g}$: extra count variability.

Priors and hyperparameters.

- $\hat{N}$: expected cells per spot. Estimate from tissue images when possible.
- α^y : how strongly y_s is kept near the batch average. Larger means tighter; smaller means more flexible.

Main assumptions. Spots are mixtures of cell-type signatures; counts are noisy negative-binomial counts; reference signatures are useful; technical variation can be represented by the model terms; related locations can share information.

Source note

This tutorial explains the Bayesian model described in the Methods section, especially page 12, of Kleshchevnikov et al., *Cell2location maps fine-grained cell types in spatial transcriptomics*, *Nature Biotechnology* 40, 661–671 (2022). The tutorial is written as an educational paraphrase with simplified examples and diagrams. The new prior-distribution section also uses the public cell2location documentation and source-code math description to explain the internal Gamma and Exponential prior structure in beginner-friendly language.